ECONOMIC FEASIBILITY STUDY OF A GPON BILLING SYSTEM

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SUPERVISOR: DR.P.O.AKUON

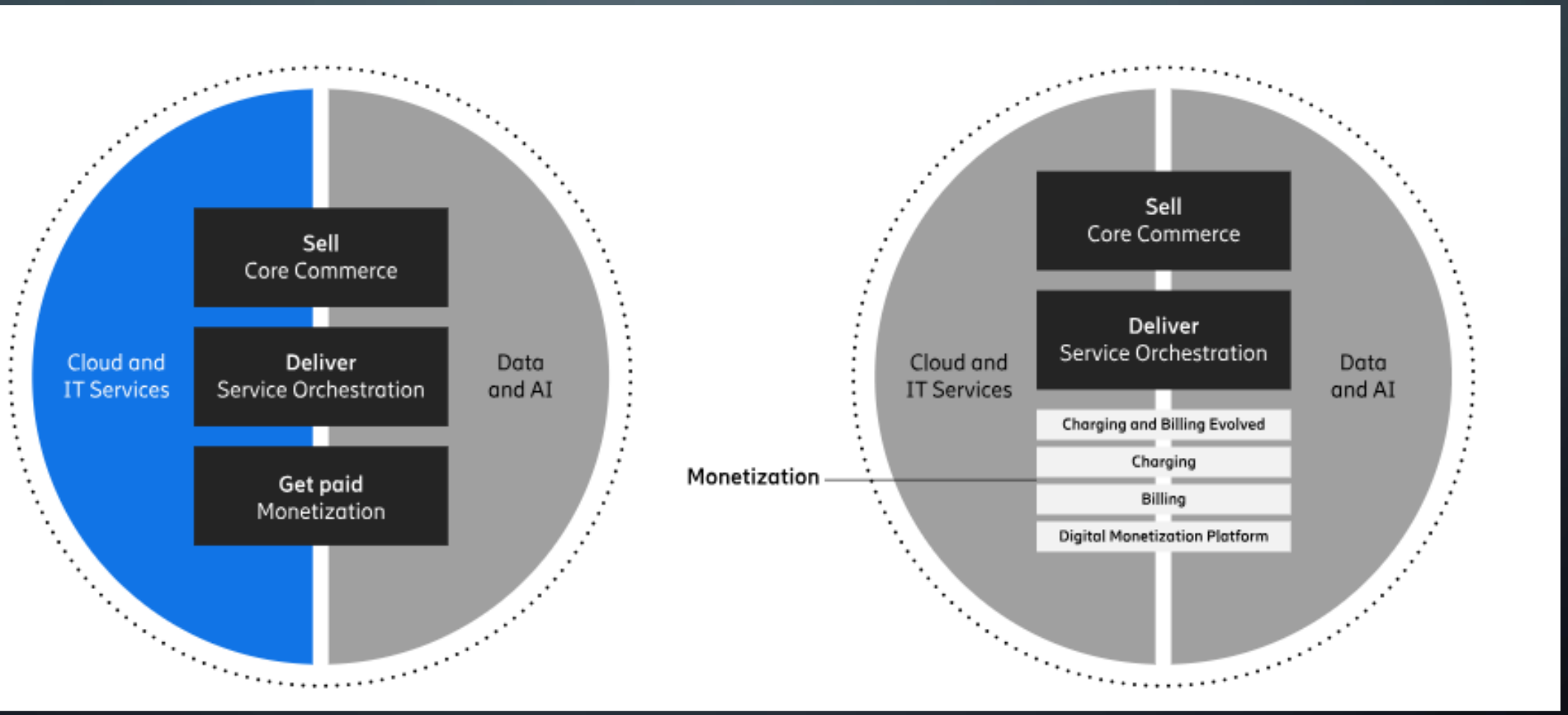
EXAMINER:DR..G.OMONDI

OUTLINE

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SECTION 3: LITERATURE REVIEW
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SECTION 5: DESIGN APPROACH
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BACKGROUND

- In an ever-growing market the need For an affordable , scalable and reliable source of internet access is becoming more necessary.
- This project in a nutshell relies on the benefits of GPON ; Gigabit Passive Optic Network a type of broadband technology that both offers multipoint connectivity to users , cheap in operation costs while retaining connectivity.
- In-order to create a system that can offer multipoint connectivity while retaining the reliability of GPON in public areas we need to implement an architecture that can handle this specifications


Sell
Core Commerce
Deliver
Service Orchestration
Get paid
Monetization
Cloud and
IT Services
Data
and AI
Monetization
Sell
Core Commerce
Deliver
Service Orchestration
Charging and Billing Evolved
Charging
Billing
Digital Monetization Platform
Cloud and
IT Services
Data
and AI

BACKGROUND

- The system architecture employed 2 parts. The BSS – Business Support Server offers billing system's intergraton and the OSS Operations Support Server that offers network support.
- Each server offers specialized tasks so that it can offer a unified approach to business operations ensuring efficiency and reliability
- But ideally that was how the project was to implemented but some processes needed more automation

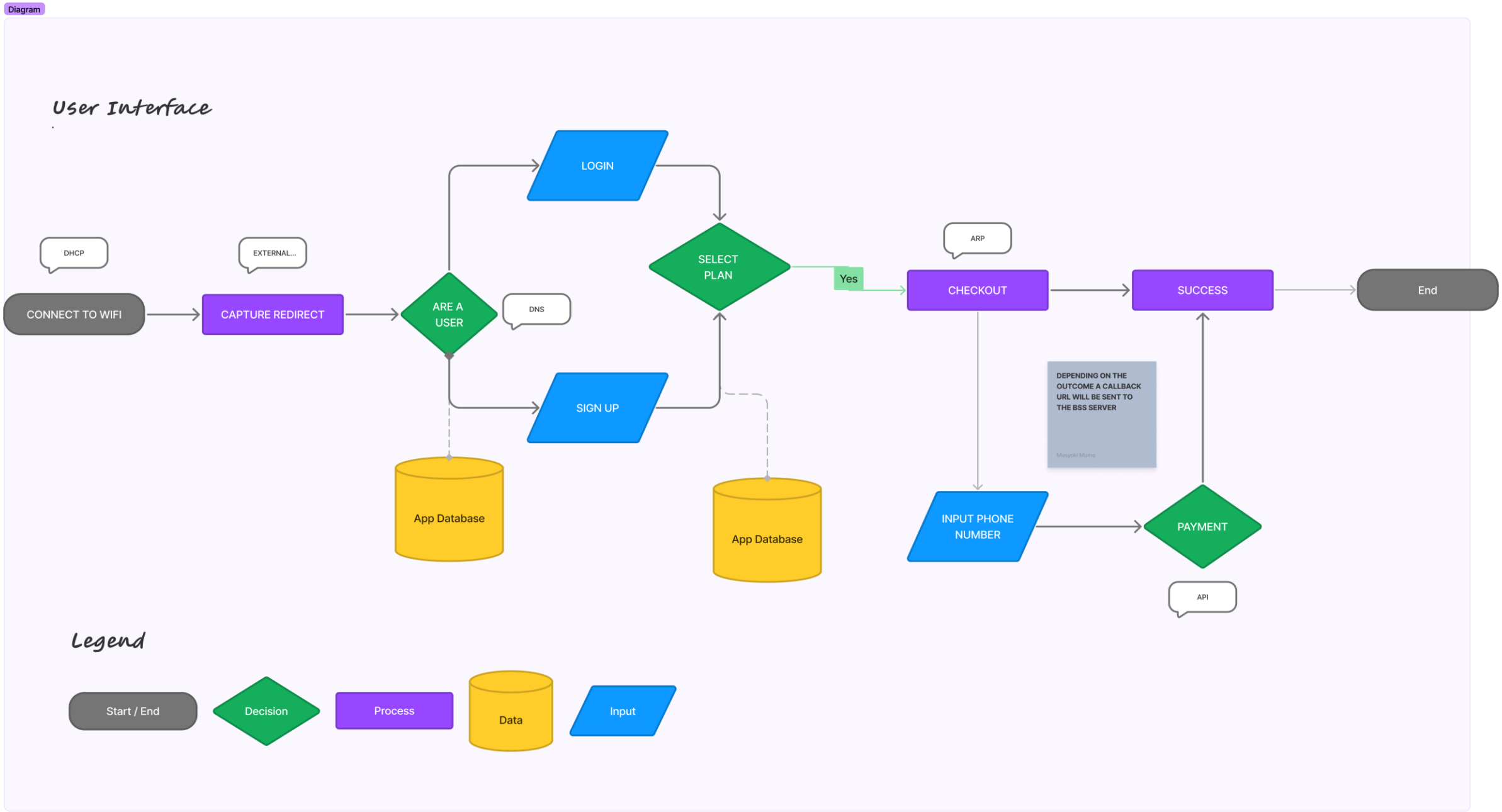
GAP / PROBLEM SOLVED

- The main issue in my system architecture was the fact that I couldn't get an operational cloud key so I had to find alternatives to it :
  - using a raspberry pi to create a network monitoring system , buring / subscribing
  - BUYING / SUBSCRIBING TO A UNIFI NETWORK MONITORING SYSTEM
  - AUTOMATING THE USER AUTHORISATION PROCESS

DESIGN APPROACH

- To achieve an affordable system I chose to use the authorization authentication route so an appropriate language employing OOP (OBJECT ORIENTED PROGRAMMING LANGUAGE) model was chosen
- The setup chosen , OS – Operational system chosen to implement the system was LINUX ; UBUNTU - DEBIAN an operational system that comes with an embedded OOP Language
- And with those in mind an appropriate UI (USER INTERFACE) model was thus devised

Diagram


User Interface
LOGIN
DHCP
EXTERNAL...
SELECT PLAN
ARP
Yes
CONNECT TO WIFI
CAPTURE REDIRECT
ARE A USER
DNS
CHECKOUT
SUCCESS
End
SIGN UP
DEPENDING ON THE OUTCOME A CALLBACK URL WILL BE SENT TO THE BSS SERVER
Musyoki Mumo
App Database
App Database
INPUT PHONE NUMBER
PAYMENT
API
Legend
Start / End
Decision
Process
Data
Input

METHODOLOGY

- To archive the system above I implemented a physical OSS layer that will be a physical implementation : Unifi Access Point &Edge Switch router
- A software layer , the BSS server and an automated system integrated between the two services whose role is to authorize purchases made on the Unifi Network Applications ; Connect the users to WiFi
- Finally the program was deployed into an independent cloud server so as to make it fully stand alone

PROGRAM DESIGN

- A key part of the program was finding an appropriate billing system
- Two options were floated Daraja API or Payhero API
- Daraja Api is a propriety of Safaricom but getting access to it's propriety usage proved challenging as an individual entity
- So the second approach was successfully integrated
- With a payment feature selected we can now move on to the implementation

BILL OF MATERIAL

	GPON BILLING SYSTEM				
SUPPLY , TEST AND COMMISSION THE FOLLOWING :					
ITEM	DESCRIPTION	UNIT	QUANTITY	RATE	TOTAL
1	Ubiquiti UniFi AC Outdoor Mesh Pro Access Point (UAP-AC-M-PRO)	NO	1	39,000	39,000
2	Ubiquiti EdgeRouter ER-6P Gigabit Router with SFP	NO	1	51,000	51,000
3	Ubiquiti UniFi CloudKey Plus SSD (UCK-G2-SSD)	NO	1	52,000	52,000
4	CAT6 CABLE AS 23 AWG (solid copper)4 Unshielded Twisted Pairs (UTP) or Shielded (FTP/SFTP) with a longitudinal separator/spline.	LM	305	LOT	25,000
5	Data router as Tenda F3 N300 300Mbps Wireless Router or approved alternative	No	1	1,500	7,500
6	Data Socket for RJ45 As SKU TR5443Or CRABTREE OR APPROVED ALTERNATIVES	NO	5	1200	6,000
TOTAL AMOUNT CARRIED FORWARD Ksh					141,500

BILL OF MATERIAL FOR THE IMPLEMENTED DESIGN

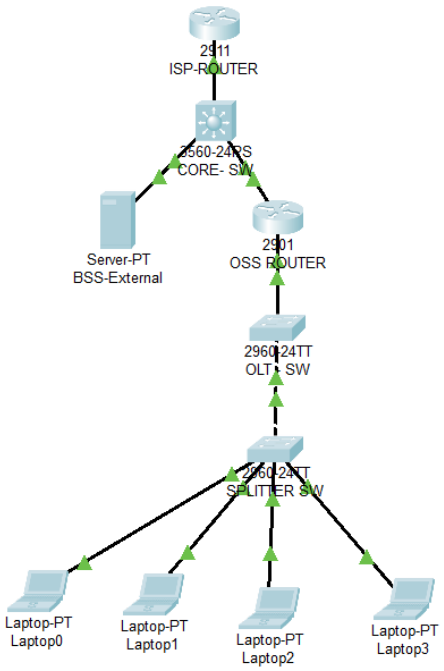
BILL OF MATERIAL

BILL OF MATERIAL WITH ADDITIONAL GATEWAY SWITCH

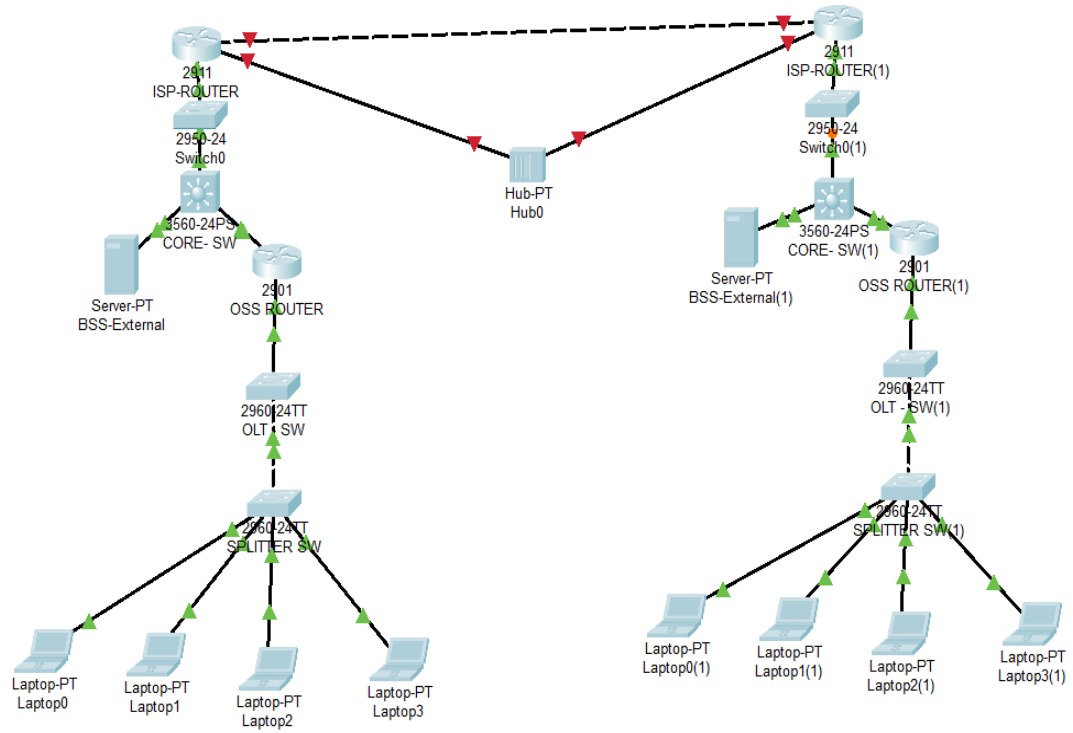
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5	Data router as Tenda F3 N300 300Mbps Wireless Router or approved alternative	No	1	1,500	7,500
6	Ubiquiti UniFi Security Gateway Pro (USG-PRO-4)		1	50,000	50,000
7	Data Socket for RJ45 As SKU TR5443Or CRABTREE OR APPROVED ALTERNATIVES	NO	5	1200	6,000
TOTAL AMOUNT CARRIED FORWARD Ksh					191,500

IMPLEMENTATION

IMPLEMENTED SYSTEM ARCHITECTURE



IMPLEMENTED SYSTEM ARCHITECTURE WITH A SECURITY GATEWAY AND REDUNDANT ISP CONNECTION WITH BGP PROTOCOL



FINANCIAL RESULTS AND DISCUSSION

- With both design sin mind , and each with independent bill of materials ; a practical feasibility study was done

  3 Case Scenarios were taken ; The International Students Accommodation at the University of Nairobi, The KMA centre at Upperhill Mara Road and a Social Gathering E.g SOLFEST 2025.
- In the next slides we will go through all the scenarios , Initial costs, expected costs and expected ROI .

FINANCIAL RESULTS AND DISCUSSION

CASE SCENARIO : LABORATORY WIFI PROTOTYPE

- A case study was done on the aforementioned area for a period of 5hours and from a sample size of 15 recurrent and concurrent users ; concurrent so that we can test scalability of the system whether it can handle all of those transactions and from that small sample an accrued amount of 245 shillings was gotten and from that with a small transaction fee from my payment proxy of 2.5 % thus the total amount earned 239 shillings.
- From these analytics we can thus extrapolate the total amount of money the system can earn in a week and thus we can get appropriate representations of the total monthly costs and the plan distribution charts – which plan gets purchased the most
- After assessing the total costs the prototype incurs if it's deployed locally , a standalone system i.e deployed on a secure webserver would mitigate the costs of electricity but we would also incur a subscription fee for the system to be constantly and optimally running

Admin Dashboard

Manage Plans | Sign Out

TOTAL DEVICES	ACTIVE SESSIONS	TOTAL REVENUE	TODAY'S REVENUE	PENDING PAYMENTS
11	0	KES 260	KES 230	5
Unique MACs	Currently online	All time	Since midnight UTC	Awaiting confirmation

Active WiFi Users (0)

No active sessions right now.

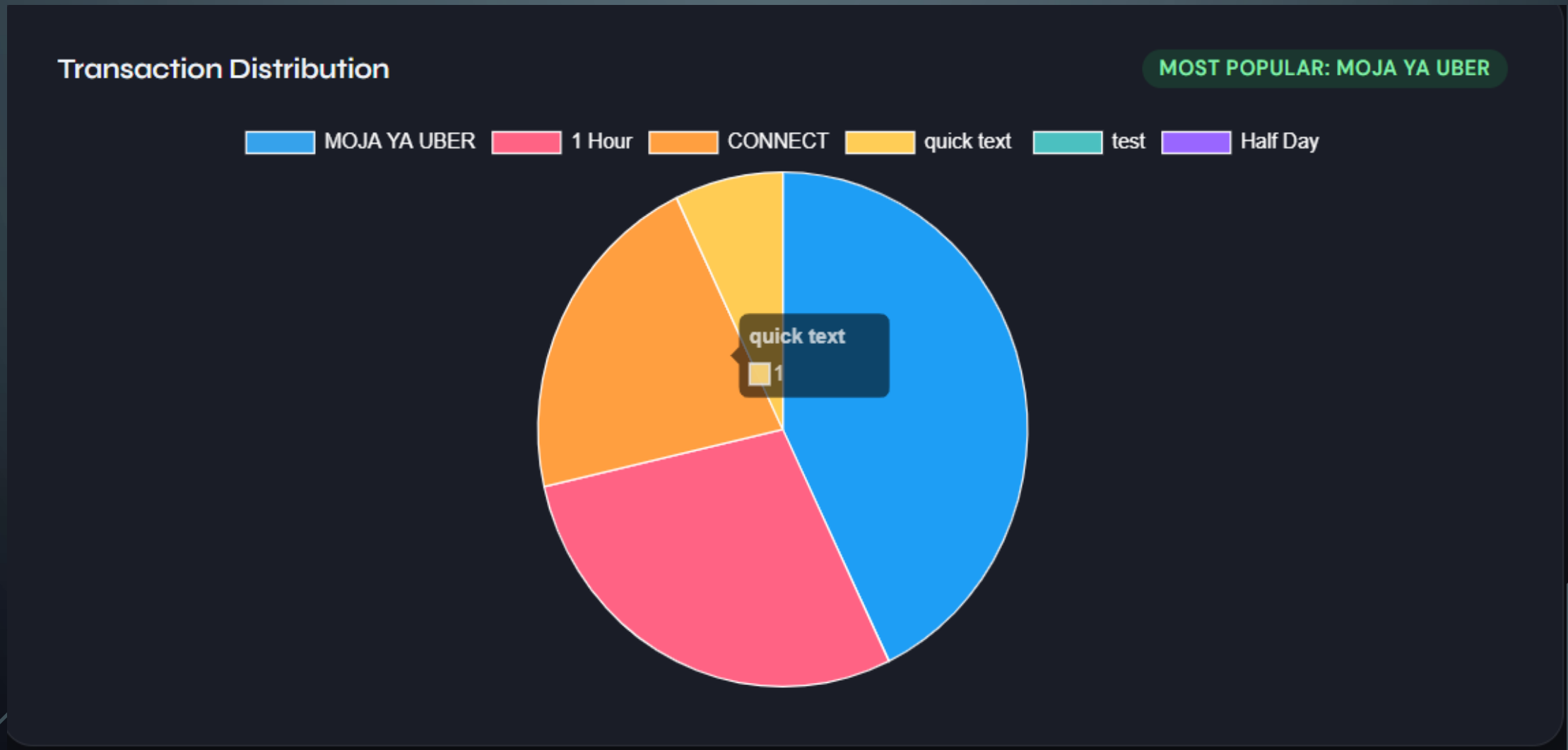
Transactions

21 shown

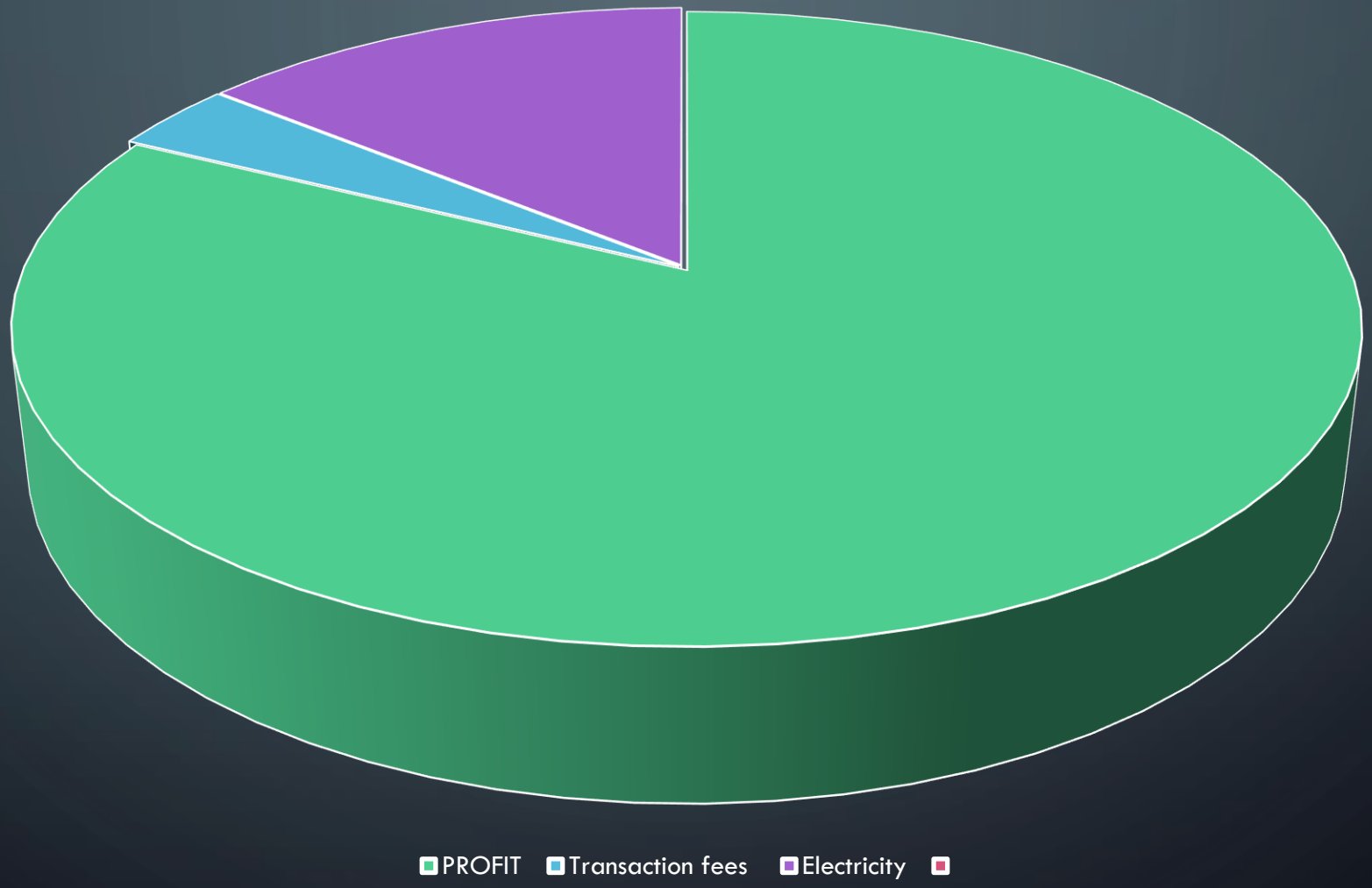
Search phone... | All statuses | mm/dd/yyyy | Filter | Clear

Reference	MAC	Phone	Plan	Amount	Status	Date
650f0c86-73b…	5e:00:6e:38:21:37	254720343790	MOJA YA UBER	KES 5	SUCCESS	2026-05-06 14:45
149dd6e4-a8f…	32:39:6d:f1:fd:96	254720343790	MOJA YA UBER	KES 5	FAILED	2026-05-06 14:44
e11eed2c-187…	6e:8d:d6:21:86:36	254724741184	MOJA YA UBER	KES 5	SUCCESS	2026-05-06 14:17
cf22bdd7-a5b…	3e:a3:b7:1b:60:f5	254742103538	1 Hour	KES 10	SUCCESS	2026-05-06 14:04

FINANCIAL RESULTS AND DISCUSSION



CUMMULATIVE FISCAL ANALYSIS



FUTURE RECOMMENDATIONS

- The only recommendation is to build a fully standalone , startup based Economic Feasibility Study for a roaming / moving GPON service exploring some of the concepts explained such the automation of the Client Authorization as well as the omission of the payment proxy and Maximizing the venture's return on investments

REFERENCES

[1] ERICSSON. (2023). OSS AND BSS: ENABLING THE DIGITAL TRANSFORMATION OF TELECOM OPERATIONS. ERICSSON TECHNICAL DOCUMENTATION. [ARCHITECTURAL DEFINITIONS ADAPTED FOR ACADEMIC CONTEXT]